

Sakhinala Sanjay Bhargav

+91 9346481321 — sanjaysakhinala@gmail.com — [linkedin.com/in/sanjay-sakhinala](https://www.linkedin.com/in/sanjay-sakhinala) — github.com/DNSdecoded
sanjaysakhinala.pages.dev — Pithapuram, Andhra Pradesh

Summary — AI/ML Engineer transforming research into production-ready solutions. Delivered 65% cost reduction in EM simulation through reinforcement learning innovation. Built and deployed multilingual NLP systems, computer vision pipelines, and secure communication platforms. Published research combining deep learning expertise with practical MLOps skills to create scalable, high-impact AI applications.

Skills

Machine Learning	Supervised & Unsupervised Learning, Deep Learning, Computer Vision, Reinforcement Learning (SAC, PPO), Transformers, Retrieval-Augmented Generation (RAG), LLMs, Generative AI	MLOps & Deployment	Docker, FastAPI, CI/CD, Model Serving, GPU Optimization, REST APIs, Vertex AI, On-Prem Deployment
ML Frameworks	PyTorch, TensorFlow, Scikit-Learn, HuggingFace, TorchVision, Stable-Baselines3, OpenCV	Data Engineering	Pandas, NumPy, SQL, BigQuery, Feature Engineering, ETL, Data Wrangling, Analytics
NLP	Multilingual NLP, Transformer Fine-Tuning, Reranking, OCR, Semantic Search	Visualization	Matplotlib, Seaborn
		Systems & Security	Python, Linux (CLI), Cryptography (AES/RSA), Networking (TCP/IP, DNS), Shell Scripting

Projects

Uncertainty-Aware RL System for EM Structure Optimization

Private Repository

- Architected reinforcement learning framework automating antenna geometry design, reducing CST simulation calls by 65%.
- Implemented uncertainty-aware Soft Actor-Critic agent with LightGBM + MLP surrogate models and custom Gymnasium environment.
- Designed fabrication-feasible reward shaping and hybrid surrogate approach with iterative CST validation, achieving convergence in 40% fewer episodes.
- **Tech:** PyTorch, CUDA, Stable-Baselines3, LightGBM, Optuna, CST API.

IndicRAG — Multilingual Indic Document QA System

GitHub: IndicRAG

- Built production-ready RAG pipeline supporting 12+ Indian languages with multilingual dense embeddings and cross-encoder reranking.
- Integrated OCR for scanned PDFs (Tesseract + LayoutParser) and semantic chunking for long documents.
- Deployed scalable FastAPI service with FAISS-based vector retrieval and mT5/mBERT models for accurate contextual answers.
- **Tech:** HuggingFace Transformers, FAISS, PyTesseract, FastAPI, RAG architecture.

CipherChat — Secure End-to-End Encrypted Messenger

GitHub: CipherChat

- Engineered CLI-based encrypted messaging system using AES-256 + RSA-2048 hybrid cryptography.
- Implemented secure key exchange, encrypted TCP sockets with SHA-256 integrity verification, and multi-user real-time messaging.
- Ensured zero-knowledge architecture with identity authentication and forward secrecy.
- **Tech:** Python, Socket Programming, Cryptography, TCP/IP, Multi-threading.

EuroSAT Land Use Classification

GitHub: landuse-cnn-eurosat

- Developed CNN achieving 89% test accuracy across 10 land-use classes on satellite imagery.
- Integrated Grad-CAM explainability for feature attribution and model interpretability.
- **Tech:** PyTorch, TorchVision, OpenCV, Grad-CAM.

DNS Resolver Performance Benchmark

GitHub: dns.analyzer

- Built multithreaded DNS benchmarking tool measuring latency across 15+ global providers.
- Automated CSV analytics and visualization for resolver performance comparison and optimization insights.
- **Tech:** Python, dnspython, Pandas, Matplotlib, concurrent processing.

Experience

- Google — AI/ML Internship (Virtual via Eduskills)

Jan 2025 – Mar 2025

AI/ML Intern

- Developed computer vision modules using Python and OpenCV to automate image detection tasks for mobile applications.
 - Evaluated model performance and deployed ML pipelines using Vertex AI and model deployment APIs.
 - Integrated ML features into end-to-end app workflows, enabling real-time inference and improved automation efficiency.
- Palo Alto Networks — Cybersecurity Internship (Virtual via Eduskills)

Oct 2024 – Dec 2024

Cybersecurity Intern

- Implemented firewall policy optimization to strengthen network access control and reduce alert false positives.
 - Analyzed threat logs and worked with SIEM workflows to identify anomalies using intelligent log-correlation systems.
 - Gained working knowledge of secure network design, intrusion detection fundamentals, and endpoint protection frameworks.
- Altair — Data Science Applied Training (Virtual via Eduskills)

Apr 2024 – Jun 2024

Data Science Trainee

- Built no-code ML pipelines in RapidMiner for data preprocessing, feature engineering, and model evaluation.
 - Applied supervised learning algorithms such as SVM and Random Forest to classification tasks and performance benchmarking.
 - Performed exploratory data analysis (EDA) and interpreted results to derive actionable business insights.

Education

- B.Tech in Electronics and Communication Engineering

May 2025

Bapatla Engineering College, Bapatla, Andhra Pradesh
- Diploma in Electronics and Communication Engineering

Oct 2021

Aditya College of Engineering, Surampalem, Andhra Pradesh
- Secondary School Certificate (SSC)

Apr 2018

Ravindra Bharathi High School, Pithapuram, Andhra Pradesh

Certifications

- Google Data Analytics Professional Certificate — Completed Mar 2025
SQL, R Programming, Data Cleaning, Data Visualization (Tableau), Business Communication, Spreadsheet Analytics
- Google Cybersecurity Professional Certificate — Completed May 2024
Network Security, Python for Security, Linux Essentials, Cloud Security, Network Architecture
- Google Prompting Essentials — Completed May 2025
Prompt engineering for GenAI workflows
- AI For Everyone — DeepLearning.AI — Completed Feb 2025
AI strategy, ML fundamentals, business use-cases
- Google Skill Badges (10+) — Completed Mar 2025
TensorFlow, Image Classification, Product Search, Object Detection, ML Learning Paths, Cloud Innovator Program, Google I/O Participation
- NPTEL Certifications — Completed Mar 2025
Cloud Computing (architecture and virtualization concepts); Computer Networks & Internet Protocols (TCP/IP, routing, DNS, network performance)

Patents

- Design of Hexagonal Patch Antenna at 28 GHz
Inventor: Sakhinala Sanjay Bhargav
Application No.: 202541014595 A — Published: Mar 7, 2025 (India)

Publications

- Bandwidth Enhancement of Slotted Hexagonal Patch Antenna for 6G Ultra-Fast Data Transfer and Brain-Computer Interface
Presented at ICMOCE 2025, IIT Bhubaneswar
DOI: 10.1109/ICMOCE64100.2025.11076991 — Publication Date: July 17, 2025
- Bandwidth Optimization of Slotted Circular Patch Antenna for 6G Ultra-Fast Data Transfer and Brain-Computer Interface Application
Presented at 16th ICCCNT 2025, IIT Indore
Proceedings submitted to IEEE Xplore — Publication Pending
- Uncertainty-Aware Reinforcement Learning System with Blended Surrogate Models for Electromagnetic Structure Optimization
Status: Under Review